

PROGRAM 2

OBJECT DETECTION USING YOLOv8

OBJECTIVE

To perform Object Detection using the YOLOv8 model and identify objects present in an image.

In this program students will learn:

- ✓ Loading YOLO Model (.pt file)
- ✓ Reading Images
- ✓ Performing Object Detection
- ✓ Drawing Bounding Boxes
- ✓ Displaying Class Labels
- ✓ Displaying Confidence Scores
- ✓ Saving Detection Results

MODEL FILE REQUIRED

Models/

└─ yolov8n.pt

DATASET STRUCTURE

Road_Images/

└─ image1.jpg

└─ image2.jpg

└─ image3.jpg

└─ ...

Any image can be used for testing.

STEP 1

INSTALL ULTRALYTICS

Purpose:

Install YOLO library.

Code:

```
!pip install ultralytics
```

Explanation:

YOLOv8 is provided by the Ultralytics package.

This step is required only once.

STEP 2

IMPORT LIBRARIES

Purpose:

Import YOLO package.

Code:

```
from ultralytics import YOLO

print("YOLO Imported Successfully")
```

Explanation:

YOLO class is used to load and run detection models.

STEP 3

LOAD YOLO MODEL

Purpose:

Load pretrained model from local folder.

Code:

```
model = YOLO(
    "Models/yolov8n.pt"
)

print("YOLO Model Loaded")
```

Explanation:

YOLOv8n

n = Nano Version

Fastest version

Suitable for lab examinations

STEP 4

LOAD IMAGE

Purpose:

Select image for detection.

Code:

```
image_path = "Road_Images/image1.jpg"
```

```
print("Image Selected")
```

STEP 5

PERFORM OBJECT DETECTION

Purpose:

Detect all objects present in image.

Code:

```
results = model(image_path)
```

```
print("Detection Completed")
```

Explanation:

YOLO automatically:

- ✓ Finds objects
- ✓ Predicts class labels
- ✓ Creates bounding boxes
- ✓ Generates confidence scores

STEP 6

DISPLAY DETECTED IMAGE

Purpose:

View detected objects.

Code:

```
results[0].show()
```

Explanation:

Displays image with:

Bounding Boxes

Class Names

Confidence Scores

STEP 7

SAVE DETECTED OUTPUT

Purpose:

Save detection result.

Code:

```
results[0].save(  
    filename="output.jpg"  
)
```

```
print("Output Saved")
```

Explanation:

Creates a new image containing:

Bounding Boxes

Object Labels

Confidence Scores

STEP 8

DISPLAY DETECTED OBJECTS

Purpose:

Print all detected classes.

Code:

```
for box in results[0].boxes:  
    class_id = int(box.cls)
```

```
print(
    results[0].names[class_id]
)
```

Example Output:

car
person
truck

STEP 9

DISPLAY CONFIDENCE SCORES

Purpose:

Check prediction confidence.

Code:

```
for box in results[0].boxes:

    confidence = float(box.conf)

    print(
        "Confidence:",
        round(confidence, 3)
    )
```

Example Output:

Confidence: 0.923

Confidence: 0.876

Confidence: 0.954

STEP 10

DISPLAY BOUNDING BOX COORDINATES

Purpose:

Locate object positions.

Code:

```
for box in results[0].boxes:

    print(box.xyxy)
```

Explanation:

Coordinates represent:

x1

y1

x2

y2

Bounding Box Corners

STEP 11

COUNT NUMBER OF OBJECTS

Purpose:

Find total objects detected.

Code:

```
count = len(
    results[0].boxes
)

print(
    "Total Objects:",
    count
)
```

STEP 12

DETECT MULTIPLE IMAGES

Purpose:

Process entire folder.

Code:

```
import os

folder = "Road_Images"

for image in os.listdir(folder):

    path = os.path.join(
        folder,
        image
    )
```

```
result = model(path)

print(
    image,
    "Processed"
)
```

IMPORTANT YOLO CONCEPTS

Bounding Box

Rectangle around object.

Class Label

Object name.

Examples:

Car

Dog

Person

Confidence Score

Probability that prediction is correct.

Higher confidence = Better prediction.

WHAT STUDENTS LEARN

✓ YOLOv8

✓ Loading .pt Files

✓ Object Detection

✓ Bounding Boxes

✓ Confidence Scores

✓ Object Localization

✓ Multiple Object Detection

✓ Saving Outputs

✓ Counting Objects

✓ Reading Images

EXPECTED OUTPUT

Input Image

↓

YOLO Model

↓

Bounding Boxes

↓

Class Labels

↓

Confidence Scores

↓

Output Image Saved

EXPECTED VIVA QUESTIONS

1. What is Object Detection?
2. Difference between Classification and Detection?
3. What is a Bounding Box?
4. What is Object Localization?
5. What is a Confidence Score?
6. Why is YOLO faster than traditional methods?
7. What is Non-Maximum Suppression (NMS)?
8. What happens if confidence threshold is increased?
9. How are multiple objects detected?
10. What are the advantages of YOLO?